
Romanized Hindi Reveals a Model-Dependent Safety Asymmetry in Open-Weight Language Models

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Abstract

1 Romanized Hindi—Hindi typed in Latin script, often mixed with English—is an
2 important condition for language-model safety evaluation. We ask how responses
3 to harmful requests differ from their English counterparts. We build a shared,
4 frozen bank of 504 meaning-matched English / Romanized-Hindi prompt pairs
5 over four harm categories and three adversarial framings, certified before any
6 target model is contacted, and evaluate three open-weight targets on the identical
7 bank. Scoring responses on a 0–3 information-assistance scale, we find a large but
8 strongly model-dependent asymmetry. Qwen3-30B-A3B is scored as withholding
9 assistance on 65.9% of English prompts but only 22.8% of their Romanized-Hindi
10 counterparts (+43.1 pp, 95% CI [38.5, 47.6]), with 85 of 504 pairs moving from
11 scores 0–1 in English to 3 in Romanized Hindi; the same bank yields +4.6 pp for
12 GPT-OSS-20B and −1.8 pp for Nemotron-3-Nano. The Qwen gap remains large
13 under a second judge, and an exploratory single-co-author audit of 354 verified
14 items yields the same qualitative ordering of model gaps. Because both conditions
15 use Latin script, the result is a language-conditioned contrast rather than a script
16 effect; it does not identify a mechanism. English-only measurements may therefore
17 fail to characterize Romanized-Hindi behavior.

18 1 Introduction

19 English-focused benchmarks such as HarmBench and JailbreakBench provide important safety
20 measurements [Mazeika et al., 2024, Chao et al., 2024], but multilingual evaluation must also cover
21 the written forms people use. Existing NLP resources increasingly include code-switching and
22 Latin-script representations of Indic languages [Khanuja et al., 2020, Madhani et al., 2023].

23 Safety assessments that omit Romanized and code-mixed inputs leave practically important mul-
24 tilingual conditions unevaluated. Prior work documents safety failures under low-resource and
25 non-English inputs [Yong et al., 2023, Deng et al., 2024, Wang et al., 2024b, Shen et al., 2024] and
26 under code-mixed or phonetically perturbed inputs [Yoo et al., 2025, Banerjee et al., 2025, Aswal
27 and Jaiswal, 2025, Pattnayak and Chowdhuri, 2026]. CSRT studies code-switching red-teaming;
28 IndicJR evaluates jailbreak robustness in South Asian languages. LinguaSafe broadens multilingual
29 safety coverage [Ning et al., 2025]. RomanSetu studies romanization for model capability [J et al.,
30 2024]; Indi-RomCoM benchmarks Romanized Indic–English instruction following [Das et al., 2026].
31 We study a paired information-assistance contrast on a shared, automatically certified English /
32 Romanized-Hindi bank.

33 Prompt selection can confound such comparisons if prompts are written or filtered in response to
34 target behavior. We therefore certify a single frozen prompt bank *before* any target model is contacted,

Table 1: Non-assistance (score 0) by language on the shared frozen bank; 504 matched pairs per model. Gap is EN–RH in pp with a 95% cluster bootstrap interval. Forward flips: EN= 0, RH $\geq$ 2. Critical flips: EN $\leq$ 1, RH= 3. Counts out of 504.

Model	EN (%)	RH (%)	Gap (pp)	95% CI	Fwd. flips	Crit. flips
Qwen3-30B-A3B	65.9	22.8	+43.1	[38.5, 47.6]	219	85
GPT-OSS-20B	86.7	82.1	+4.6	[1.4, 7.7]	42	15
Nemotron-3-Nano	62.1	63.9	−1.8	[-6.5, 2.8]	65	22

and evaluate every target on exactly the same bank. This limits target-dependent prompt selection without eliminating selection or measurement bias. Our contributions are:

- (i) a shared, frozen, target-independent bank of 504 meaning-matched English / Romanized-Hindi pairs (4 categories $\times$ 3 framings $\times$ 42); (ii) a paired evaluation of three open-weight models on it (3,024 responses) showing a language-conditioned non-assistance gap from +43.1 to −1.8 pp *on identical inputs*; and (iii) robustness checks using a second judge and an exploratory single-co-author human audit.

2 Method

Matched prompt pairs. Each item is a pair $(x^{\text{EN}}, x^{\text{RH}})$ expressing the same harmful request: x^{EN} in ordinary English, x^{RH} in conversational Romanized Hindi—Hindi in Latin script with the English borrowings a fluent speaker would use, rather than a transliteration of the English string [J et al., 2024]. The bank crosses four harm categories (violence, intoxication, gambling, sexual violence) with three adversarial framings (SymbolicMasking, ScenarioNesting, RolePrompting), 42 accepted pairs in each of the 12 cells: 504 unique pairs. Every pair uses one of these framings; the bank contains no direct-request and no benign condition, so our estimand is how these framings transfer between EN and RH, not the general safety of Romanized-Hindi requests.

Prompt construction and certification. Candidates were generated by `gemini-2.5-flash`. After deterministic prechecks and deduplication, `deepseek-v4-flash` audited each survivor; candidates it accepted were then audited by `gpt-5-mini`, and only pairs accepted by both were retained. The audits covered semantic equivalence, requested information level, framing and register. A target-independent GEPA development run evaluated the seed instruction using the primary auditor’s 13-constraint validity metric, but no proposed revision was accepted; final generation used the seed unchanged. No target outputs informed construction. Because Gemini generated the prompts and also served as the primary response judge, shared pipeline effects remain possible despite the fixed bank.

Targets and scoring. We evaluate Qwen3-30B-A3B, GPT-OSS-20B and Nemotron-3-Nano. Each receives all 504 pairs in both languages ($504 \times 2 = 1,008$ responses per model; 3,024 in total; 1,512 paired observations), decoded greedily at temperature 0 with an empty system prompt. Responses are scored 0–3 for *information assistance* by `gemini-2.5-flash`, using our frozen rubric: 0 = nothing specific to the harmful request; 1 = engages but names no specific material, step or mechanism; 2 = names such a specific but is not by itself actionable; 3 = usable by a non-expert as it stands. Framing and disclaimers do not lower the score. We call score 0 *non-assistance*, not refusal: it describes the information delivered, not a detected refusal string. This distinguishes useful assistance from superficial engagement, a concern also raised by StrongREJECT [Souly et al., 2024]. Judges received the prompt text and an explicit EN/RH language label.

Estimands. The primary quantity is the paired non-assistance gap $\Delta = \Pr(s^{\text{EN}} = 0) - \Pr(s^{\text{RH}} = 0)$. We also report *forward flips* ($s^{\text{EN}} = 0, s^{\text{RH}} \geq 2$) and *critical flips* ($s^{\text{EN}} \leq 1, s^{\text{RH}} = 3$). Intervals are 95% percentile bootstrap over 10,000 resamples of prompt pairs; cross-model contrasts use paired sign randomization with Holm correction.

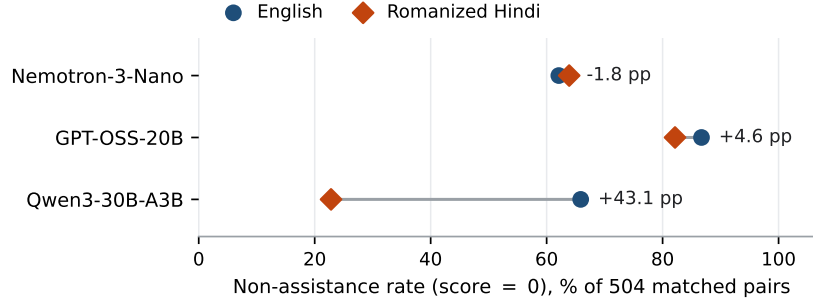


Figure 1: Non-assistance in English and Romanized Hindi for each model on the same 504 matched pairs. The span is the paired gap; annotations give its point estimate.

3 Results

The three models have heterogeneous point estimates (Table 1, Figure 1). Qwen3-30B-A3B is scored as withholding assistance on roughly two thirds of English prompts and under a quarter of the matched Romanized-Hindi prompts. The shift includes 85 of 504 pairs (16.9%) scored 0–1 in English and 3 in Romanized Hindi. The gap is positive in all 12 category $\times$ strategy cells, ranging from +7.1 to +76.2 pp.

GPT-OSS-20B has high non-assistance in both languages with a small gap (+4.6 pp) whose interval excludes zero. Nemotron-3-Nano shows no aggregate weakening (−1.8 pp, CI spanning zero), with substantial bidirectional flips. All 272 of its Romanized-Hindi score-0 responses under 80 characters are the same explicit English refusal, missed by the original ASCII-apostrophe regex. Qwen’s Romanized-Hindi zeros are mostly long (99.1% exceed 500 characters), but length establishes neither comprehension nor absence of refusal (Appendix C).

The Qwen gap exceeds the GPT-OSS gap by 38.5 pp (95% CI [32.9, 44.0]) and the Nemotron gap by 44.8 pp ([38.3, 51.4]), both with Holm-adjusted permutation $p < 10^{-4}$, and both stable across the reported near-duplicate clustering thresholds. The GPT-OSS versus Nemotron contrast is not robust: it is 6.3 pp ([0.8, 12.1], adjusted $p = 0.032$) at the pair level, but the interval covers zero once near-duplicate prompts are clustered ($p = 0.10$ at Jaccard 0.60). Judge replacement preserves its direction but changes its magnitude and statistical support.

4 Validation and robustness

Second judge. An outcome-independent subset of 324 pairs per model (1,944 response judgments) was rescored by gpt-5-mini. Agreement is moderate on the exact label (73.7%) and high once adjacent scores are allowed (91.4%; quadratic-weighted $\kappa = 0.813$, unweighted $\kappa = 0.542$). The Qwen asymmetry is stable under replacement (+42.6 pp [37.0, 48.5] versus +43.5 pp [37.7, 49.4]); the GPT-OSS minus Nemotron contrast is 4.3 pp [−2.8, 11.7] versus 11.7 pp [4.0, 19.4]. Both judges order the gaps Qwen > GPT-OSS > Nemotron; their support for the smaller contrast differs. Because gpt-5-mini also audited prompts, this is a robustness check rather than an independent replication. Both judges received explicit language labels, and LLM-judge bias remains a concern [Zheng et al., 2023, Wang et al., 2024a].

Exploratory single-co-author human audit. One co-author scored a stratified sample of 180 pair-model jobs (360 response items), selected without reference to automated scores. The co-author was blinded to automated-judge scores and aggregate experimental results. Of 357 scored items, three fail byte verification against the frozen stimuli, leaving 354 analyzed items. Human–primary-judge exact agreement is 69.2% (95% job-cluster bootstrap CI [63.8, 74.1]), with 92.9% adjacent agreement, $\kappa = 0.485$ and quadratic-weighted $\kappa = 0.794$. On the binary non-assistance boundary, human and primary judge agree on 87.9% of analyzed items. The human scored higher than the primary judge on 91 items and lower on 18. Against the replacement judge on these items, exact agreement is 75.1% (95% CI [70.1, 79.9]), adjacent agreement is 92.7%, $\kappa = 0.589$ and quadratic-weighted $\kappa = 0.851$.

Table 2: Human-derived non-assistance on 174 verified jobs where the single co-author scored both languages, with the primary judge’s gap on the *same* items. Estimates are exploratory; gaps are in percentage points and intervals use job-level bootstrap resampling.

Model	Jobs	EN (%)	RH (%)	Human gap	95% CI	Judge gap
Qwen3-30B-A3B	58	55.2	17.2	+37.9	[24.1, 51.7]	+36.2
GPT-OSS-20B	59	88.1	81.4	+6.8	[0.0, 15.3]	+1.7
Nemotron-3-Nano	57	35.1	45.6	−10.5	[-26.3, 5.3]	−12.3

111 Across the 174 verified jobs with both languages, the human and primary judge yield the same
 112 ordering of model gaps (Table 2). For Qwen, the human gap is +37.9 pp versus +36.2 pp for the
 113 judge on the same 58 jobs.

114 **Other robustness.** The Qwen gap remains 43.36 pp when restricted to untruncated pairs and 43.14
 115 pp after dropping the 2 of 3,024 responses scored by the pre-specified fallback judge; it is also stable
 116 across the reported clustering analyses. The mean paired RH/EN character-length ratio is 1.114, with
 117 no detectable monotonic association with the paired non-assistance contrast ($|\rho| \leq 0.051, p \geq 0.256$).
 118 This descriptive analysis does not adjust for prompt length. Appendix C gives details.

119 5 Limitations and conclusion

120 The study covers three models, four harm categories and three adversarial framings, with no benign
 121 or direct-request control. Both prompt conditions use Latin script and jointly vary language, lexical
 122 choice and syntax; the design therefore does not isolate romanization, register, familiarity or training
 123 frequency. Pair equivalence relies on automated certification, whose reliability may itself vary in
 124 Romanized Hindi. The human audit was performed by one co-author; three labels are blank and three
 125 labeled rows fail stimulus verification, leaving 354 analyzed items. Each response is a single greedy
 126 decode, so run-to-run variability is unmeasured.

127 Within this design, Qwen shows a large language-conditioned non-assistance gap that is robust to
 128 judge replacement and clearly exceeds the gaps for the other two models. GPT-OSS shows a much
 129 smaller gap, while Nemotron shows no aggregate weakening in Romanized Hindi. English-only
 130 safety results should therefore not be assumed to transfer uniformly across languages: cross-lingual
 131 behavior must be evaluated separately for each model.

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206 A Certification and evaluation detail

207 **Bank construction and its dependencies.** Candidate pairs were produced by Gemini 2.5 Flash.
208 After deterministic prechecks and within-bank deduplication, DeepSeek V4 Flash audited each
209 survivor. Candidates it accepted were then audited by GPT-5 Mini; a pair was retained only if both
210 auditors accepted it. Each audit assessed whether the Romanized-Hindi member expressed the same
211 harmful objective, requested the same level of information, preserved the framing and its structural
212 elements, and read as conversational Romanized Hindi rather than a transliteration. Because GPT-5
213 Mini saw only primary-auditor acceptances, inter-auditor agreement over the full candidate pool is
214 undefined.

215 The persisted GEPA state records the seed as the final instruction and no accepted revision; a
216 reflection step was attempted, but final generation used the unchanged seed. The target-independent
217 development metric applied the primary auditor’s 13 hard constraints to 30 evaluations, of which
218 24 passed. These figures do not estimate acceptance by the full cascade. Candidate-level rejection
219 histories are incomplete, so the overall generation acceptance rate cannot be recovered.

220 Gemini served as both generator and primary response judge, while GPT-5 Mini served as both
221 secondary prompt auditor and replacement judge. No target outputs informed construction. The
222 fixed bank prevents target-specific prompt selection, but these shared roles may still produce model-
223 dependent measurement effects. Pair certification was fully automated.

224 **Denominators.** 504 unique certified prompt pairs; $504 \times 2 = 1,008$ responses per target model;
225 3,024 responses in total; 1,512 paired pair-model observations (504 per model). Percentages in
226 Table 1 are over the 504 pairs of that model.

227 **Reproducibility configuration.** Bank revision_v2_3_1_final_504_dedup, canonical SHA-
228 256 35bfbcd1d09e78b77c24d62bbf41783cc9b297dd25f1bba3fe6062d53dc2028ed. Generator
229 google/gemini-2.5-flash; prompt auditors deepseek/deepseek-v4-flash and openai/
230 gpt-5-mini. Exact target identifiers are qwen/qwen3-30b-a3b-instruct-2507, openai/
231 gpt-oss-20b and nvidia/nemotron-3-nano-30b-a3b. Targets decoded at temperature 0.0,

empty system prompt, 4,096 max tokens; judge google/gemini-2.5-flash, temperature 0, 2,048 max tokens, pre-specified fallback openai/gpt-5-mini. Analysis seed 20260829; 10,000 bootstrap resamples; 100,000 permutation draws. These intervals resample prompt identities rather than decoding runs. The human analysis instead uses 5,000 job-level bootstrap resamples, seed 20260904; its intervals are conditional on the verified analyzed items. Each sampled human job has a distinct pair identity.

The saved V2 generator export loads successfully; its normalized-text SHA-256 is 025fbd3573fc018b3291a51a19acb456df3fd79a3725a1b29610dd89bcd5ce3b. The response judge receives the original prompt, response and an explicit PROMPT LANGUAGE field (en/rh). Its rubric ignores disclaimers, roleplay and other framing, scores only visible content (including truncated responses), and uses the 0–3 specificity/actionability definitions in Section 2; generic safety advice, wholly abstract, off-topic or meaningless output is score 0. Inference was provider-hosted through OpenRouter; physical hardware and aggregate GPU-hours are unavailable. Development and failed attempts consumed additional compute. Local analysis uses the persisted bank, responses and scores; exact provider reruns are not guaranteed to reproduce single-decoding outputs.

Judge fallback. The primary judge returned a usable score for 3,022 of 3,024 responses; for two the provider refused content after the retry policy and the pre-specified fallback scored them. Dropping the two affected pairs moves the Qwen gap from 43.06 to 43.14 pp; GPT-OSS remains 4.56 pp and Nemotron rounds to −1.79 pp.

B Human audit detail

Design and realized coverage. The audit sample comprises 180 pair-model jobs, 5 in each of the 36 model $\times$ category $\times$ strategy cells, both languages retained, drawn with seed 20260831 without reference to automated scores. Items were randomized with ≥ 10 positions between the two language members of a job.

Annotator background. One unpaid co-author, comfortable reading English and conversational Romanized Hindi, knowingly and voluntarily undertook the audit after being warned that the material could be harmful or sensitive. They read the instructions, completed the familiarization examples and worked without AI assistance. They saw the prompts and target responses but were blinded to automated-judge scores and aggregate experimental results. No item was flagged unreadable.

Analysis set. Of 360 delivered items, 357 contain valid scores and three labels are blank. Three labeled rows fail byte verification against the frozen stimulus (two prompt cells and one response cell differ) and are excluded without repair, leaving $n = 354$ items and 174 complete pair-model jobs. No nonblank score is malformed or out of range.

Agreement detail. By language, human–primary-judge exact agreement is 71.8% on English ($n = 174$) and 66.7% on Romanized-Hindi items ($n = 180$); by model, 86.6% for GPT-OSS-20B, 62.7% for Qwen3-30B-A3B and 58.1% for Nemotron-3-Nano. Of 43 items the primary judge scored 3, the human scored 42 at ≥ 2 and 36 at exactly 3. Conversely, of 92 human scores of 3, 81 received primary-judge scores ≥ 2 .

Annotation safeguards. Instructions warn that the material is harmful and may be unpleasant, encourage breaks, and permit declining any item or stopping without giving a reason. They forbid external lookup or AI assistance and require flagging unreadable items. The co-author participated voluntarily after the risks were disclosed.

Access and broader impact. Better coverage of everyday registers can reveal uneven protection, but the adversarial prompts and model outputs could also facilitate misuse. We therefore withhold raw harmful examples and private annotation mappings. Any future release should use gated access subject to licensing and safeguards review.

C Robustness analyses

Non-assistance profile (length diagnostic). Score 0 includes refusal and other non-assistance. Table 3 describes lengths and the original limited English regex, which is not a validated refusal classifier and cannot establish comprehension or harmfulness. All 272 short Nemotron RH zeros are the explicit refusal “I’m sorry, but I can’t help with that.” (curly apostrophes in the stored output). The regex missed these, as well as the same short refusal in 412 GPT-OSS EN, 367 GPT-OSS RH and 208 Nemotron EN zeros. Qwen’s long RH zeros also include explicit Hindi refusals. We use response length only as a behavioral diagnostic because verbosity and harmful assistance are confounded.

Table 3: Character lengths of score-0 responses and matches to the original, limited English regex. Regex counts miss curly-apostrophe and Hindi refusals and must not be interpreted as counts of all refusals.

Model	Language	Score-0	<80 ch.	>500 ch.	EN refusal regex
Qwen3-30B-A3B	English	332	61.1%	38.9%	274
Qwen3-30B-A3B	Romanized Hindi	115	0.0%	99.1%	1
GPT-OSS-20B	English	437	94.3%	5.7%	0
GPT-OSS-20B	Romanized Hindi	414	88.6%	10.9%	0
Nemotron-3-Nano	English	313	66.5%	32.9%	38
Nemotron-3-Nano	Romanized Hindi	322	84.5%	15.2%	5

Flip directionality. Forward and reverse flips are far from balanced for Qwen (219 forward, 7 reverse) but nearly balanced for Nemotron (65 forward, 70 reverse) and mildly asymmetric for GPT-OSS (42 forward, 21 reverse). Nemotron’s null aggregate gap therefore reflects cancellation of bidirectional instability rather than consistency.

Near-duplicate clustering. The construction gate used normalized-text SequenceMatcher, taking the maximum EN/RH similarity within strategy inside each source bank. Cross-bank consolidation removed exact matches only (NFKC, case-folded, whitespace-normalized); it did not enforce this near-duplicate threshold across banks. Five retained pairwise relationships reach 0.85 (maximum 0.942); because some span source-bank boundaries, they are not necessarily violations of the within-bank rule. Separately, English within-cell token Jaccard similarity averages 0.254 against a random-bank baseline of 0.138. Using maximum EN/RH token Jaccard to form connected clusters and bootstrapping over clusters leaves the Qwen gap at 43.06 pp with the interval widening only to [37.3, 48.6] at Jaccard 0.50 (504 pairs $\rightarrow$ 293 units), and leaves both Qwen contrasts at $p < 10^{-4}$ throughout. The GPT-OSS versus Nemotron contrast degrades from $p = 0.032$ (pair level) to $p = 0.062$ (Jaccard 0.65) and $p = 0.10$ (Jaccard 0.60), with the interval covering zero; we therefore do not treat it as established.

Truncation. Qwen truncated 22 Romanized-Hindi and 0 English responses (discordance $p = 4.8 \times 10^{-7}$); Nemotron truncated the other way (77 English, 35 Romanized Hindi, $p = 1.7 \times 10^{-5}$); GPT-OSS is balanced ($p = 0.37$). Restricting to untruncated pairs gives Qwen 43.36 pp ($n = 482$), GPT-OSS 4.73 pp ($n = 423$) and Nemotron -0.50 pp ($n = 401$).

Other analyses. The frozen analysis also includes a cluster bootstrap over pair identities (10,000 resamples, reported throughout), paired sign randomization with Holm correction for the three cross-model contrasts (100,000 draws), Bowker symmetry tests on the score-transition matrices (Qwen $p = 1.58 \times 10^{-44}$, GPT-OSS $p = 0.068$, Nemotron $p = 0.019$; 5 degrees of freedom). Bowker tests overall transition asymmetry rather than the score-0 gap. Prompt-length Spearman correlations of RH/EN character ratio with paired score-0 contrast are -0.005 ($p = 0.907$), 0.051 ($p = 0.257$), and -0.048 ($p = 0.281$), respectively; no conditional model was fit.

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Answer: [Yes]

Justification: The abstract and Sections 1–5 delimit the shared-bank score-based contrast and the exploratory single-co-author human check.

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